

Kinora Imaging & Clinical Decision Rules

Clinical AI Orientation (Part 19)

XR/MRI/CT/US Indications, Ottawa, Canadian C-Spine, NEXUS

Version 1.0 — Kinora Admin Conocimientos Upload

Disclaimer

Educational only. Imaging decisions follow clinical rules and local protocols.

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1. Modality Indications

RECORD: X-ray

ID: IMG-01

Indications: Trauma fracture/dislocation, OA baseline, Ottawa-positive ankle/knee, Canadian C-Spine positive

Limits: Poor early stress Fx/soft tissue

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: MRI

ID: IMG-02

Indications: Disc/nerve/cord, ligament/tendon, occult Fx/stress, infection/tumor soft tissue, myelopathy

Limits: Cost/claustrophobia; incidental findings common

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: CT

ID: IMG-03

Indications: Complex fractures, surgical planning, dens/Jefferson detail, when MRI contraindicated for bone

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Ultrasound

ID: IMG-04

Indications: Tendon dynamic, Baker cyst vs DVT adjunct pathway, infant hip, guided injection

Limits: Operator dependent; deep joints limited

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

2. Clinical Decision Rules

RECORD: Ottawa Ankle/Foot Rules

ID: IMG-CDR-01

Purpose: Reduce unnecessary ankle/foot XR after acute injury

Criteria: Malleolar/midfoot pain PLUS bone tenderness OR inability to WB 4 steps

Evidence: High sensitivity validated

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Ottawa Knee Rules

ID: IMG-CDR-02

Purpose: Acute knee trauma XR decision

Criteria: Age $\geq$ 55, isolated patellar tenderness, fibular head tenderness, inability flex 90°, inability WB 4 steps

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Canadian C-Spine Rule

ID: IMG-CDR-03

Purpose: Alert trauma patients — who needs c-spine imaging

High_Risk: Age $\geq$ 65, dangerous mechanism, paresthesias → image

Low_Risk_Then_ROM: If low-risk factors allow safe assessment, rotate 45° each side — if unable, image

References: Canadian C-Spine Rule literature

RECORD: NEXUS Criteria

ID: IMG-CDR-04

Purpose: C-spine imaging decision (alternative framework)

Criteria: No midline tenderness, no focal neuro, normal alertness, no intoxication, no painful distracting injury — if all met, imaging may be deferred

References: NEXUS literature

3. Radiological Findings (Orientation)

RECORD: Common Findings Glossary

ID: IMG-FIND-01

Spine: Disc herniation, Modic, stenosis, listhesis, OPLL

Shoulder: Full-thickness RCT, labral tear — correlate clinically

Knee: ACL bone bruise pattern, meniscal signal grades

Ankle: Mortise clear space, syndesmosis diastasis

Principle: Image findings must match clinical presentation — avoid treating MRI alone

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.